Bridge Day 5 INSTRUCTOR KEY

Compound Boolean Evidence and Complete Decisions

Friday, July 10, 2026 • 20 points • target 16 minutes

Name		UIN		Section	

Assessment boundary and evidence source

Contract: 07a04826c975
Cumulative foundations: input conversion, comparisons, and/or/not, branch order, f-strings
Scaffold level: complete compound-condition program
Evidence source: the matching v5 workbook readiness/evidence pages and VIP activities 18.13, 18.14, 18.15, 18.16, 18.17.
Show intermediate state for trace credit. Program credit requires one coherent solution, a stated assumption when the prompt leaves one implicit, and at least one test whose expected result can distinguish correct from incorrect logic.

Item 1. Compare two Boolean groupings. - 5 points

```
trained = False
temperature = 22
override = True

ready = trained and 15 <= temperature <= 30 or override
safe = trained and (15 <= temperature <= 30 or override)
print(ready, safe)
```

Question: Evaluate each component and explain why the two stored results differ.

Model response: Python evaluates and before or. ready becomes (False and True) or True, which is True. safe requires trained on the outside, so False and (True or True) is False. Exact output: True False.

Item 2. Trace negation and a dependent readiness rule. - 5 points

```
flow_ok = True
label = "pending"
needs_review = not flow_ok or label != "approved"
ready = flow_ok and not needs_review
print(needs_review, ready)
```

Question: Evaluate each Boolean in dependency order and give exact output.

Model response: not flow_ok is False, but label != approved is True, so needs_review is True. ready then combines True with not True, which is False. Exact output: True False.

Complete program - 10 points

- Read flow as a float, approval as text, and sensor as text.
- Print invalid if flow is negative.

- Otherwise print ready only when flow is from 18 through 22 inclusive, approval is exactly yes, and sensor is exactly ready.
- Print hold for every other nonnegative case. Print exactly one word.

Program workspace

Model program

```
flow = float(input())
approval = input()
sensor = input()

if flow < 0:
    print("invalid")
elif 18 <= flow <= 22 and approval == "yes" and sensor == "ready":
    print("ready")
else:
    print("hold")
```

Test input, expected result, and why this test matters

Reasoning and test evidence The invalid-input branch comes first. The ready branch names all three required pieces of evidence with and, so no single passing condition can approve the record. Protective tests include 18/yes/ready and 22/yes/ready as ready; 22.1/yes/ready and 20/no/ready as hold; and -1/yes/ready as invalid.

Scoring guidance: 2 input/conversion, 2 invalid guard, 3 complete compound condition, 2 exclusive exact output, 1 boundary tests.

Quiz evidence is sampled from the taught construct boundary; workbook and notebook evidence remain the primary learning surfaces.